PLAYGROUND DESIGN PROPOSAL

SELAMBER PRIMARY SCHOOL

SITE PARAMETERS

School	6 of 6
Total Students	68
Compound Area	24,333 m²
Play Zone	≈ 160 m²
Buildings	8 (1,460 m²)
Open Space	22,873 m²

EXISTING EQUIPMENT

- 2 Slides — Functional
- 1 Swing — Functional
- 2 Ladders — Functional
- 1 Balance Beam — Functional

EQUIPMENT GAPS

- 1 Merry-Go-Round — Nonfunctional (Remove)
- See-Saw — Missing
- Climbing Structure — Missing

Site analysis, concept design, and detailed design proposal for pre-primary playground infrastructure serving 68 children aged 5–6 years within a 24,333 m² school compound. The pre-primary play zone of approximately 160 m² is a dedicated section within the larger primary school campus, designed to provide a safe, stimulating play environment separate from older students. This smaller school follows the Minimum Viable / Phased approach (Option B) to optimise resource allocation.

Prepared for: Cities Alliance & Assosa City Education Office
Implementing Partner: International Rescue Committee (IRC)

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PHASE 1 — SITE ANALYSIS

1.1 Site Context Report

Selamber Primary School is situated within Assosa City, the capital of Benishangul-Gumuz Regional State in western Ethiopia. The school compound occupies a **24,333 m² parcel**, making it the largest of the six SUIDAC programme school sites by total area. Eight permanent FIAER-type buildings with a combined footprint of **1,460 m²** are arranged in a loose cluster within the compound, leaving an expansive **22,873 m² of available open space**. Despite the large compound size, the pre-primary section serves only **68 students** (30 male, 38 female) aged approximately 5 to 6 years, classifying it as a smaller school within the programme.

The compound is bounded by dense residential development to the west, with the Plan International office located to the southwest and the IRC office nearby to the southeast. A corrugated metal fence encloses the compound perimeter and has been assessed as structurally adequate — it will be retained without replacement. Trees along the boundary provide partial shade and have been marked for preservation as ecological and shade assets. The school is located within an urban neighbourhood with good road access, facilitating material delivery and construction supervision.

The existing play equipment inventory is relatively well-stocked for the small student population. Two functional slides, one functional swing, two functional ladders, and one functional balance beam are currently installed. However, one **nonfunctional merry-go-round** has been assessed as structurally unsafe and will be removed prior to any new construction. Three critical equipment categories are absent from the site: a see-saw, a climbing structure, and a tunnel crawl. The tunnel crawl is deferred to Phase 2 under the phased implementation approach.

The site currently suffers from **overgrown tall grass throughout the compound**, which must be cleared before any construction activity begins. An old metal arch and climbing frame showing significant rust corrosion has been identified for removal. Discarded tires are scattered across the ground; these present both a tripping hazard and a mosquito breeding risk. Where possible, intact tires will be recycled as border edging for play zones or pathway delineation. The ground surface is uneven and will require grading to achieve the minimum 1.0 percent slope for adequate drainage before any equipment installation can proceed.

Climate conditions at Assosa are characterised by a mean annual temperature of approximately 22 °C, with daily maxima occasionally exceeding 35 °C during the long dry season (October through May). Annual rainfall averages 1,200 mm, concentrated in the kiremt rainy season from June to September. These conditions necessitate robust shade provision within all play zones and a gravity-based drainage system to prevent standing water accumulation, which poses both safety and health risks for young children.

1.2 Constraint Map Description

The site has been divided into four colour-coded constraint zones based on the overlay analysis of the KML boundary data, VLM image interpretation, and field observations. These zones guide all subsequent layout decisions and equipment placement within the large compound.

Zone	Colour	Description	Area (m²)	Constraint
RED — No-Go	Red	Fence setback zone (1.5 m from fence line), building wall setbacks (3.0 m for swings/slides). Includes nonfunctional merry-go-round removal zone and rusted metal arch/climbing frame removal zone.	≈ 520	Hard exclusion

Zone	Colour	Description	Area (m^2)	Constraint
YELLOW — Caution	Amber	Existing equipment zones requiring retrofit inspection, uneven ground areas requiring grading, mature boundary tree root protection zones (1 m radius from trunk). Discarded tire scatter zones.	≈ 380	Conditional use
GREEN — Available	Green	Open cleared areas between buildings, existing play space identified in VLM analysis. Primary targets for new equipment installation and phased expansion.	≈ 4,200	Preferred
BLUE — Infrastructure	Blue	Pathway corridors (1.2 m min width), drainage swales, and utility connection points. Must remain clear for maintenance access and water flow.	≈ 1,100	Service reserve

Table 1. Constraint zone classification and area allocation.

1.3 Opportunity Assessment

Selamber School presents a distinctive opportunity profile: the compound area (24,333 m^2) is the largest in the programme, yet the pre-primary population (68 students) is the smallest. This creates an exceptionally favourable space-to-child ratio. The active play area requirement is 80 m^2 per 50 students, scaled proportionally: 80 × (68/50) = **108.8 m^2 minimum**. With an additional 20 percent quiet/shade zone allocation of approximately **27 m^2**, the total minimum play infrastructure area is approximately **136 m^2**. This is well within the identified available open space, leaving substantial room for generous equipment spacing, wide safety zones, and clearly marked Phase 2 expansion areas.

The boundary trees provide partial shade that will reduce surface temperatures under play equipment by an estimated 6–10 °C during peak afternoon heat. While canopy coverage is not as dense as some other school sites, these preserved trees can be supplemented with additional plantings along the quiet zone perimeter to expand shade coverage progressively as new trees mature. The proximity to the IRC office and Plan International office provides logistical advantages for supervision visits, material storage, and rapid response to any construction issues.

The existing functional equipment (2 slides, 1 swing, 2 ladders, 1 balance beam) provides a solid foundation that can be retained, refurbished with anti-rust treatment (zinc-chromate primer and enamel topcoat), and incorporated into the new layout. This significantly reduces Phase 1 procurement costs and shortens the construction timeline. The removal of the nonfunctional merry-go-round and the rusted metal arch will free approximately 25 m^2 of space for new equipment installation. The discarded tires, where structurally intact, offer a free material resource for creative border edging and pathway delineation, reducing procurement needs.

PHASE 2 — CONCEPT DESIGN

2.1 Design Criteria and Scaling

All concept designs have been developed in accordance with the non-negotiable design standards specified in the IRC SUIDAC Playground Design Manual. Equipment dimensions, safety zone clearances, surface treatment depths, and material specifications follow the prescribed values exactly. The following scaling calculations apply to Selamber Primary School's pre-primary enrolment of 68 students:

- **Active Play Zone:** 80 m² per 50 students × (68/50) = **108.8 m² ≈ 110 m²**
- **Quiet/Shade Zone:** 20% of total play area = 0.20 × 136 ≈ **27 m²**
- **Total Play Infrastructure Area:** ≈ **136 m²** (within 160 m² dedicated zone)
- **Pathways:** Estimated 45 m × 1.2 m = **54 m²**
- **Minimum Slides Required:** 1 per 50 students × (68/50) = **1.36 → 2 slides** (existing retained)
- **Minimum Swings Required:** 1 per 30 students × (68/30) = **2.27 → 3 seats** (1 existing + 1 new bay)

> Material Treatment Requirements: All metal components shall receive zinc-chromate primer followed by enamel topcoat (2 coats). All timber and bamboo shall undergo boron-based anti-termite treatment via 24-hour soak before installation. River sand surfacing shall be washed and graded to 150–300 mm depth under all equipment. Pathways shall use fine gravel at 100 mm compacted depth.

2.2 Option A — Maximum Play Value (Reference Only)

Option A represents the theoretical maximum play value configuration for a school of this size, providing the full complement of equipment types specified in the SUIDAC manual. This option includes two slides, three swing seats across two bays, one merry-go-round (replacement), two see-saws, two climbing structures, two tunnel crawls, two balance beams, and two sets of stepping logs — totalling 16 individual installations across 8 equipment categories. While this option provides the most comprehensive play experience, it represents an over-investment for a school serving only 68 children. The total equipment surface area under Option A would be approximately 168 m², exceeding the minimum active play zone requirement by 53 percent.

The principal disadvantage of Option A for Selamber is the ratio of equipment density to student population. With only 68 pre-primary students, the full complement of equipment would result in underutilisation and higher per-child maintenance costs. The merry-go-round replacement alone costs approximately ETB 12,000 and requires a 4.3 m diameter safety zone, consuming space disproportionate to its play value for a small cohort. For these reasons, Option A is presented here as a reference for future Phase 2 expansion but is **not recommended as the initial implementation strategy**.

ID	Equipment	Qty	Status	Surface Area (m²)
E1	Merry-Go-Round	1	New replacement	16.8
E2	See-Saw	2	New	44.0
E3	Slide	2	Existing retained	21.0
E4	Chain Swing	2 bays (3 seats)	1 existing + 1 new bay	35.3
L1	Balance Beam	2	Existing retained	15.7

ID	Equipment	Qty	Status	Surface Area (m²)
L2	Climbing Structure	2	New (bamboo dome)	19.2
L3	Tunnel Crawl	2	New	13.5
L4	Stepping Logs	2 sets	New	11.5

Table 2. Option A equipment schedule — Maximum Play Value (reference only).

2.3 Option B — Minimum Viable / Phased (SELECTED)

Given the small student population of 68 children, **Option B: Minimum Viable / Phased** has been selected as the recommended design for Selamber Primary School. This option provides the essential core equipment needed to deliver a comprehensive developmental play experience while optimising resource allocation. Phase 1 retains all functional existing equipment (2 slides, 1 swing, 2 ladders, 1 balance beam) and adds only two critical missing pieces: one new see-saw (E2) and one new bamboo climbing structure (L2). The nonfunctional merry-go-round is removed, and the tunnel crawl and stepping logs are deferred to Phase 2 expansion.

The total Phase 1 equipment surface area is approximately **82 m²** of washed river sand, which, combined with interstitial circulation space, provides an effective active play area of approximately 110 m² — meeting the scaled minimum requirement. This compact layout is concentrated adjacent to the pre-primary classroom to minimise supervision distances and maximise the use of partial shade from nearby boundary trees. The Phase 2 expansion zone is clearly demarcated with low timber bollards and surfaced with compacted fine gravel, maintaining usability as informal play space in the interim period.

The pathway network under Option B consists of a single 1.5 m wide main path connecting the pre-primary classroom entrance to the active play zone, extending to the school gate for access. A secondary 1.2 m path connects the active zone to the quiet/shade area under the boundary trees. Drainage follows the same gravity-based principle as all SUIDAC designs, with a shallow swale along the main path alignment directing surface water toward the fence line collection point. The total estimated cost of Phase 1 is approximately 35 percent of a full Option A implementation, with Phase 2 expansion costing an additional 55 percent of the Phase 1 budget when funding becomes available.

Item	Phase 1 (Core)	Phase 2 (Expansion)	Total
Slides	2 (existing retained)	—	2
Chain Swings	1 (existing, 1 bay)	1 new bay (2 seats)	3 seats
See-Saws	1 new	1 new	2
Climbing Structure	1 new (bamboo)	1 new	2
Merry-Go-Round	Remove (nonfunctional)	— (deprioritised)	0
Balance Beams	1 (existing retained)	1 new	2
Ladders	2 (existing retained)	—	2
Tunnel Crawl	—	1 new	1
Stepping Logs	—	1 set	1 set
Quiet/Shade Zone	27 m² (under trees)	10 m² expansion	37 m²

Table 3. Phased equipment plan for Option B — Minimum Viable (Selected).

2.3.1 Phase 1 Surface Area Calculations

Each equipment piece requires a defined surface area including the equipment footprint plus the mandatory safety zone. The following calculations detail the Phase 1 surface requirements:

Equipment	Footprint (m^2)	Safety Zone (m^2)	Total Surface (m^2)	Surface Type
E3 Slide (×2, existing)	2 × 3.0 × 1.5 = 9.0	2 × 6.0 = 12.0	21.0	River sand
E4 Chain Swing (×1, existing)	6.25	12.5	18.8 (≈ 19.0)	River sand
E2 See-Saw (×1, new)	4.0 × 0.5 = 2.0	8.0 × 3.0 − 2.0 = 22.0	24.0 (≈ 24.0)	River sand
L2 Climbing Structure (×1, new)	4.0 (2.0 × 2.0)	5.6 ($\pi \times 3.5^2/4 - 4.0$)	9.6 (≈ 10.0)	River sand
L1 Balance Beam (×1, existing)	3.5 × 0.25 = 0.9	3.5 × 2.0 = 7.0	7.9 (≈ 8.0)	River sand

Table 4. Phase 1 surface area calculations per equipment type (Option B).

Total Phase 1 equipment surface area: 82.0 m^2 of washed river sand at 150–300 mm average depth. Combined with approximately 28 m^2 of interstitial circulation space, the effective active play area reaches 110 m^2, satisfying the scaled minimum requirement of 108.8 m^2.

2.3.2 Safety Zone Verification

All safety zones have been verified against the mandatory minimum clearances. The following checks confirm compliance with the IRC SUIDAC Playground Design Manual:

Check Item	Standard	Design Value	Status
Fall zone around moving equipment	≥ 2.0 m	2.0 m (swing, see-saw)	PASS
Clearance from fence line	≥ 1.5 m	1.5 m maintained	PASS
Clearance: swings/slides to buildings	≥ 3.0 m	3.5 m minimum	PASS
Pathway width	≥ 1.2 m	1.5 m main, 1.2 m secondary	PASS
Equipment height limit	≤ 2.5 m	Max 2.0 m (slide)	PASS
Safety zone overlap (inter-equipment)	None permitted	All verified clear	PASS

Table 5. Safety zone compliance verification matrix.

2.3.3 Supervision Sight-Line Analysis

Teacher supervision is a critical safety requirement for pre-primary playgrounds. The compact Option B layout concentrates all Phase 1 equipment within a 12 m × 10 m area adjacent to the pre-primary classroom. This deliberate proximity ensures that the maximum sight-line distance from the classroom entrance to any equipment piece is approximately **18 metres** — well within the recommended 30 metre maximum for unobstructed visual supervision. A single teacher stationed at the classroom doorway can observe all children on all equipment simultaneously without any blind spots.

The climbing structure (L2), which presents the highest supervision challenge due to its enclosed bamboo dome form, is positioned closest to the classroom at approximately 8 m distance. The open-grid bamboo construction ensures 360-degree visibility through the structure. Slides and swing are placed at intermediate distances of 10–15 m, while the balance beam is positioned furthest from the building at approximately 18 m but remains fully visible due to the flat topography and absence of any visual obstructions in the cleared play area.

2.3.4 Drainage Management

The drainage strategy employs a minimum 1.0 percent slope across all play zones, graded from north (near buildings) to south (toward fence line). Surface water flows via shallow swales (150 mm deep, 300 mm wide) along the pathway network toward a collection point at the southern fence line. From there, water discharges through a grated outlet into the existing compound drainage. The compact Option B layout reduces the total swale length to approximately 25 m, compared to 60 m for a full Option A implementation, lowering both construction effort and long-term maintenance requirements.

Under each equipment installation, the 150–300 mm layer of washed river sand serves as both a fall-attenuation surface and a percolation medium. The sand layer is underlain by geotextile fabric that prevents mixing with the native lateritic subsoil while allowing water to drain through freely. This dual-function approach eliminates the need for sub-surface drainage pipes and can be maintained by community labour through periodic sand loosening and top-up every 6–12 months.

PHASE 3 — DETAILED DESIGN (OPTION B SELECTED)

3.1 Master Layout Plan Description

The master layout for Selamber Primary School organises the playground into four distinct functional zones within the dedicated pre-primary play area. The layout prioritises teacher sight-lines, natural drainage, and compact spatial efficiency appropriate for a 68-child cohort, while preserving boundary trees and maintaining clear expansion zones for Phase 2.

Zone 1 — Transition Area (Building Perimeter): A 3.0 m wide transition strip surrounds the nearest pre-primary classroom buildings, providing dry, safe access between classrooms and the play areas. This strip is surfaced with compacted fine gravel (100 mm depth) over geotextile fabric. No equipment is installed within this zone. The transition area also accommodates the existing discarded tire collection point during the clearance phase; intact tires are removed, cleaned, and relocated to the pathway border edging.

Zone 2 — Active Play Area (Central): The primary play zone occupies approximately 110 m² (including circulation) in the cleared open area adjacent to the pre-primary classroom. Equipment is arranged in a compact semi-circular pattern with the climbing structure (L2) and swing (E4) positioned closest to the building for maximum supervision proximity. The two existing slides (E3) are positioned on the eastern flank with south-facing orientations. The new see-saw (E2) is placed on the western side of the active zone, oriented along the east-west axis. The existing balance beam (L1) is positioned at the southern edge of the active zone. A 1.5 m wide main pathway bisects the zone from north to south, providing both circulation and emergency access.

Zone 3 — Quiet/Shade Zone (Eastern Boundary): The preserved boundary trees along the eastern fence provide partial canopy cover for a 27 m² quiet zone. Simple log seating and a small shade platform are installed beneath the tree canopy for reading, rest, and low-activity play. This zone is separated from the active play area by a 1.2 m gravel pathway that serves as a physical and visual buffer. The quiet zone is designed for expansion in Phase 2 when additional shade structures may be added.

Zone 4 — Phase 2 Expansion Area (Southern): A clearly demarcated expansion zone of approximately 80 m² is reserved south of the active play area for future Phase 2 installations. This zone is marked with low timber bollards (400 mm height, 500 mm spacing) and surfaced with compacted fine gravel at 80 mm depth. In the interim period, it serves as informal open play space for running games and group activities. No permanent installations are permitted in this zone until Phase 2 funding is confirmed and the detailed design is updated.

3.2 Equipment Placement Schedule

The following table provides relative placement coordinates for each Phase 1 equipment installation. Coordinates are referenced from the southwestern corner of the play zone fence line (origin: 0, 0), with X increasing eastward and Y increasing northward. All positions maintain the required safety clearances from buildings, fences, and adjacent equipment.

ID	Equipment	Qty	Zone	Rel. Position (X, Y) m	Orientation	Notes
E3a	Slide (existing)	1	2	(10, 10)	S-facing	Refurbished, anti-rust treatment
E3b	Slide (existing)	1	2	(16, 10)	S-facing	Refurbished, anti-rust treatment

ID	Equipment	Qty	Zone	Rel. Position (X, Y) m	Orientation	Notes
E4	Chain Swing (existing)	1	2	(4, 12)	N/A	Existing bay, 2 seats, refurbished
E2	See-Saw (new)	1	2	(16, 6)	E-W axis	New installation, 4.0 m length
L2	Climbing Structure (new)	1	2	(4, 7)	N/A (dome)	Bamboo A-frame dome, 1.2 m H
L1	Balance Beam (existing)	1	2	(10, 3)	N-S axis	Existing, relocated if needed

Table 6. Phase 1 equipment placement schedule with relative coordinates (Option B).

3.3 Grading and Earthwork Notes

The existing ground surface across the play zone is uneven, a condition documented during the site protocol assessment. Grading is required to achieve the minimum 1.0 percent slope for drainage and to create level equipment platforms. The total earthwork volume is modest due to the compact Option B footprint.

Earthwork Item	Description	Estimated Volume	Method
Site clearing	Removal of overgrown tall grass, discarded tires, and debris from play zone and pathways	N/A (labour)	Hand tools, wheelbarrow
Hazard removal	Removal of nonfunctional merry-go-round and rusted metal arch/climbing frame	N/A (labour)	Bolt cutters, hand winch
Slope correction	Grade play zone to 1.0% fall from north to south (currently uneven, no consistent slope)	≈ 15 m^3 cut/fill	Hand-compacted laterite
Equipment platforms	Level pads under each equipment installation, 100 mm above surrounding grade for drainage	≈ 5 m^3 fill	Compacted gravel base
Drainage swale	150 mm deep × 300 mm wide swale along southern edge, 25 m length	≈ 1.2 m^3 excavation	Hand-dug, geotextile lined
Pathway sub-base	Fine gravel, 100 mm compacted over geotextile, 45 m total length	≈ 5.4 m^3	Imported gravel, hand-compacted
Sand surfacing	Washed river sand, 200 mm avg depth under all equipment zones (82 m^2)	≈ 16.4 m^3	Imported, levelled by hand
Tree protection mounds	Circular earth mounds around preserved boundary tree trunks (4 trees, 1 m radius)	≈ 1.5 m^3	Hand-shaped

Table 7. Grading and earthwork schedule.

Total estimated earthwork volume: 45 m^3 (cut/fill). All earthwork shall be executed by community labour using hand tools (shovels, rakes, hand tampers). No mechanical equipment is required. The compact Option B footprint significantly reduces earthwork compared to a full Option A implementation, keeping the project achievable within community labour capacity. Cut-and-fill balance is achieved by reusing excavated swale material as fill for slope correction and equipment platforms.

A compaction standard of 95 percent Modified Proctor Density shall be achieved for all filled areas using hand tampers. Quality assurance shall be performed by the site supervisor using a simple probe test (steel rod penetration resistance) at 3 m intervals across all graded areas. The smaller play zone footprint means fewer probe test points are required compared to larger school sites.

3.4 Planting Plan

The planting strategy focuses on three objectives: supplementary shade provision within the quiet zone, erosion control on graded surfaces, and visual enhancement of the playground environment. All species are selected for local availability within 50 km of Assosa, drought tolerance, non-toxicity to children, and low maintenance requirements. The existing boundary trees are preserved and supplemented with additional plantings.

Species (Local Name)	Type	Qty	Location	Purpose	Spacing
Ficus sycomorus (Shola)	Deciduous shade tree	2	Eastern boundary, within quiet zone	Supplementary shade canopy	8 m
Terminalia brownii (Dhidhessa)	Semi-deciduous tree	1	Southern expansion zone, adjacent to drainage swale	Root stabilisation for swale banks	—
Codiaeum variegatum (local ornamental)	Low shrub	8	Along pathway edges, alternating sides	Visual definition, windbreak	1.5 m
Cynodon dactylon (Bermuda grass)	Ground cover	80 m²	Quiet zone and expansion areas	Erosion control, dust suppression	Broadcast seed
Euphorbia tirucalli (local hedge)	Low hedge (0.4 m)	18 m length	Boundary between active and quiet zones	Visual separation, no climbing risk	0.5 m (linear)

Table 8. Planting schedule — locally sourced species.

The existing boundary trees shall be preserved in their entirety. No pruning shall be performed without arborist assessment. Root protection zones of 1.0 m radius around each trunk shall be maintained during grading and construction. Circular earth mounds shall be built around the trunks to prevent soil compaction and root damage from foot traffic. The recycled tires from the site clearance phase, where structurally intact, will be placed as border edging around the planting beds, providing both a visual demarcation and a secondary use for waste materials.

3.5 Maintenance Framework

Long-term playground sustainability depends on a clear maintenance protocol that can be executed by school staff and community volunteers with basic tools. The compact Option B layout reduces the total maintenance burden compared to larger school sites. The following framework defines maintenance tasks, frequencies, responsible parties, and material requirements.

Task	Frequency	Responsible	Materials/Tools	Notes
Sand loosening and top-up	Monthly	School groundskeeper	Rake, wheelbarrow, river sand	Check depth, replace if below 150 mm

Task	Frequency	Responsible	Materials/Tools	Notes
Metal equipment rust check	Quarterly	School maintenance lead	Wire brush, primer, enamel paint	Touch-up zinc-chromate + enamel coat
Timber/bamboo termite check	Bi-annually (pre/post rain)	Community volunteer team	Boron solution, brush	Re-soak if soft spots detected
Fastener and joint inspection	Monthly	School groundskeeper	Wrench set, replacement bolts	Check swing chains, slide anchors, see-saw pivot
Drainage swale clearing	After each rain event	School staff	Shovel, hand trowel	Remove debris, check geotextile integrity
Pathway gravel replenishment	Bi-annually	Community volunteer team	Fine gravel, rake	Top up to 100 mm, re-compact
Tree watering (new plants)	Daily for first 3 months	Student water monitors (supervised)	Watering cans	Morning watering, 5 L per tree per day
Fence integrity check	Quarterly	School maintenance lead	Pliers, wire, replacement posts	Check corrugated metal fence for gaps, rust
Grass cutting in quiet zone	Monthly (rainy season)	School groundskeeper	Sickle or machete	Maintain ground cover below 100 mm height

Table 9. Maintenance schedule and responsibility matrix.

3.6 Material Sourcing — Local Availability

All materials specified in this design are available within a 50 km radius of Assosa city. This local sourcing requirement supports the community-labour construction model and minimises procurement lead times and transportation costs. The proximity of Selamber School to the IRC office and Plan International office further facilitates logistics coordination.

Material	Source Location	Distance from Site	Estimated Cost Rate
Washed river sand	Blue Nile riverbed, south of Assosa	25 km	ETB 800/m³
Fine gravel (pathways)	Assosa municipal quarry	8 km	ETB 650/m³
Bamboo (construction grade)	Mao-Komo district forests	40 km	ETB 150/pole
Timber (Eucalyptus posts)	Assosa plantation nursery	12 km	ETB 200/post
Metal pipe (see-saw, slide)	Assosa metalworks workshop	5 km	ETB 3,500/piece
Zinc-chromate primer	Assosa hardware shops	3 km	ETB 450/gallon
Enamel topcoat paint	Assosa hardware shops	3 km	ETB 550/gallon
Boron treatment chemical	IRC supply chain (Addis Ababa)	Air freight	Included in IRC logistics
Geotextile fabric	IRC supply chain	Air freight + 5 km	Included in IRC logistics
Chain and hardware (swing)	Assosa metalworks / Bahir Dar	5–180 km	ETB 2,000/set
Timber bollards (expansion zone)	Assosa plantation nursery	12 km	ETB 150/bollard
Grass seed (Bermuda)	Assosa agricultural supply	5 km	ETB 200/kg

Table 10. Local material sourcing schedule.

3.7 Removal and Recycling Plan

Selamber School has two significant removal items that must be addressed before new construction begins: the nonfunctional merry-go-round and the rusted metal arch/climbing frame. Additionally, the scattered discarded tires require management. The following plan details the removal, disposal, or recycling approach for each item.

Item	Condition	Action	Disposal / Recycling	Estimated Effort
Merry-go-round	Nonfunctional, structurally unsafe; central bearing seized, rust perforation on platform	Full removal; cut bolts, dismantle on site	Scrap metal sold to Assosa recycler; revenue offsets labour costs	4 labour-hours, bolt cutters + hand winch
Metal arch/climbing frame	Severe rust corrosion; structural integrity compromised	Full removal; cut at base with angle grinder	Scrap metal sold to Assosa recycler	2 labour-hours, angle grinder
Discarded tires (estimate 15–20)	Mixed condition; some intact, some cracked	Sort on site: intact tires retained, damaged tires removed	Intact tires: repurposed as border edging for play zones and pathway definition. Damaged tires: Assosa municipal waste collection	6 labour-hours, hand tools
Overgrown tall grass	Full compound coverage; some areas above 1 m height	Slash, clear, and rake from play zone	Compost on-site or transport to municipal green waste. Grass roots left in quiet zone for natural regrowth where desired	12 labour-hours, sickles, rakes

Table 11. Removal and recycling plan for existing hazards.

The total estimated removal effort is **24 labour-hours**, achievable by a team of four community labourers over two working days. All scrap metal removals generate revenue through sale to the Assosa metal recycler, partially offsetting the labour cost. The recycled tire border edging eliminates the need to procure commercial rubber or timber edging, saving an estimated ETB 3,000 in material costs while diverting waste from the municipal disposal system.

End of Design Document — Selamber Primary School (School 6 of 6)

This document was prepared under the IRC SUIDAC programme for Cities Alliance donor review. All dimensions, calculations, and specifications are site-specific to Selamber Primary School and should not be applied to other school sites without independent verification.